

Traceability between HLL and Pivot Models

To ensure correct correspondence and traceability between the source (HLL) and target (Pivot) models, we adopted an approach combining Epsilon's native trace mechanism with explicit semantic identifiers.

First, we used the `.equivalent()` operation throughout the ETL rules. This guarantees that structural links in the source model are perfectly preserved in the target model.

Second, we mapped the identifiers of source elements directly onto their target equivalents, providing explicit, human-readable evidence of traceability in the generated `.xmi` files:

- **Nodes:** A Pivot Node inherits its id directly from the HLL Scene's name (`n.id = sc.name`), making every node immediately traceable to its source scene.
- **Edges:** A Pivot Edge encodes both its parent scene and the originating choice in its id (`e.id = parentScene.name + "_choice_" + ch.id`). For example, the edge `dana_room_choice_1` unambiguously documents that it was produced from choice 1 inside scene `dana_room` in the HLL model.

Together, these two mechanisms ensure that any element in the Pivot model can be traced back to its exact origin in the HLL model, both programmatically via the ETL trace and visually by inspecting the generated `.xmi`.